



Department of Computer Science and Engineering Data Science

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Experiment No. 10

Aim:- Develop an application and analyze its performance using any two machine learning classification algorithms.

Output:-

Working on dataset : wine_dataset (prebuild in sklearn)

Import Libraries :

```
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

Load Dataset :

```
wine = load_wine()
X = wine.data
y = wine.target
```



Split – 70/30 ratio ~ 30(test_size) and 70(train_size) :

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.3, random_state=42  
)
```

Scale data : to transform data to have a mean = 0, standard deviation = 1

```
scaler = StandardScaler()  
X_train = scaler.fit_transform(X_train)  
X_test = scaler.transform(X_test)
```

Fitting 1st Model : Logistic Regression with 5000 Iterations with its accuracy :

```
lr = LogisticRegression(max_iter=5000)  
lr.fit(X_train, y_train)  
  
y_pred_lr = lr.predict(X_test)  
print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))  
  
Logistic Regression Accuracy: 0.9814814814814815
```

Fitting 2nd Model :

```
dt = DecisionTreeClassifier()  
dt.fit(X_train, y_train)  
  
y_pred_dt = dt.predict(X_test)  
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))  
  
if(accuracy_score(y_test, y_pred_lr) > accuracy_score(y_test, y_pred_dt)):  
    print("Logistic Regression is better")  
else:  
    print("Decision Tree is better")  
  
Decision Tree Accuracy: 0.9629629629629629
```



Comparison of models based on accuracy :

```
if(accuracy_score(y_test, y_pred_lr) > accuracy_score(y_test, y_pred_dt)):  
    print("Logistic Regression is better")  
else:  
    print("Decision Tree is better")
```

```
Logistic Regression is better
```

Hence, Logistic Regression with accuracy $\approx 98.14\%$, Decision Tree with accuracy $\approx 96.29\%$.

Therefore, Logistic Regression is the best model to work with wine dataset.